



We are done. Go through this and tell me what we are seeing. Interpret all the results. Explain to me what has happened. What should be our next step. Go figure by figure and interpret the results and provide justification and reasoning for the same

Now I have a complete picture of the run. Here is the full interpretation, figure by figure and result by result.

What Happened: The Full Run

This was the first successful dual-channel training run of the InceptionResNet1D with `in_channels=2`, trained on the `windows_plus/` dataset containing primary transit (channel 0) + secondary eclipse (channel 1) windows. Training ran on an RTX 3070 for 985 minutes (~16.4 hours) across 80 epochs.

Dataset Sanity Check Results

The data audit confirmed everything was clean and correctly shaped:^[1]

Split	Files	Windows	Shape	Positive%	NaN/Inf
Train	47	11,791,812	(N, 2, 201)	24.1%	Clean
Val	10	1,160,105	(N, 2, 201)	18.4%	Clean
Test	10	1,161,306	(N, 2, 201)	20.5%	Clean

The two-channel shape (N, 2, 201) is confirmed throughout — the dual-channel implementation is working correctly. The training set is richer in positives (24.1%) because augmentation (noise injection and phase shifting) was applied only to training positives, inflating their count. The natural `pos_weight` computed from the training distribution is 3.15; the used `pos_weight` is 4.2, which is 33% above natural — intentionally elevated to bias toward recall.^[2]

Figure 1: Training History — Loss Curves and Metrics

What you see

Left subplot (loss): Train loss and val loss over 80 epochs. Train loss starts high (~0.76 at epoch 1) and descends steadily to settle around 0.363 by the late epochs. Val loss descends initially but shows volatile oscillation throughout, with a catastrophic spike to 3.53 at epoch 76, then recovers. The best val loss was around 0.262–0.289 in the epochs surrounding epoch 65.

Right subplot (val metrics): Precision, recall, F1, and AUC-ROC over epochs. A vertical dashed line marks the best epoch (epoch 65, AUC = 0.9429).

Interpretation

Epochs 1–2: All-Positive Collapse. Both early epochs triggered the all-positive collapse warning — precision 0.190 (below the 0.20 floor), recall ~0.998, AUC 0.477. This is expected and self-limiting: a `pos_weight` of 4.2 initialises the loss landscape heavily favouring recall, so the untrained model defaults to predicting everything as positive. The model escaped this regime on its own by epoch 3–4 as the gradient signal from true negatives accumulated.^[1]

Epochs 3–65: Healthy learning phase. AUC rises steadily from ~0.5 to 0.9429 at epoch 65. The train loss plateau around 0.363 is consistent — it signals that the training set is essentially solved at the current model capacity and regularisation level, not overfitting in the classical sense. Val loss oscillation during this period is normal given the streaming dataset's per-epoch randomisation.

Epoch 76: Catastrophic spike. Val loss explodes to 3.5303, AUC crashes to 0.1586, precision drops to 0.180, and recall returns to 1.000 — another all-positive collapse, but this time at epoch 76, well into the OneCycleLR cosine annealing tail. At this point the LR was $4.81\text{e-}08$ — nearly zero. This is the key diagnostic signal: a catastrophic loss spike at near-zero LR is *not* a learning rate issue. It means the model encountered a pathological batch (or series of batches) where the gradient, even at tiny LR, destabilised the weights enough to temporarily collapse the decision boundary. The model partially recovers to AUC ~0.938 the next epoch, confirming the weights were not destroyed, just temporarily perturbed.^[1]

Epochs 77–80: Incomplete recovery. The model recovers AUC to ~0.935–0.938 but never returns to 0.9429. At epoch 80 the threshold sweep fails entirely — "no threshold reached recall 0.85, best recall 0.496 at t=0.05" with AUC 0.8379 — indicating this final epoch produced a poorly calibrated output. The best checkpoint from epoch 65 is correctly what gets used for evaluation.^[1]

Train vs. val loss gap: Train loss ~0.363, val loss oscillating between 0.26 and 0.52 (excluding the spike). Notably, val loss is sometimes *lower* than train loss in the vicinity of the best epoch. This is a known artefact of dropout being active during training (stochastic) but disabled during eval, and of the dataset composition — val has 18.4% positives vs. train's 24.1%, making the val BCE loss structurally easier.

Figure 2: Confusion Matrix (Val Set, $t=0.69$)

	Predicted Negative	Predicted Positive
Actual Negative	TN 827,306	FP 126,784
Actual Positive	FN 14,943	TP 191,072

Interpretation

True Positives (TP): 191,072. The model correctly flags 92.75% of all transit windows. This is the single most important number for a transit survey — missed transits are permanently lost from the candidate pipeline.^[1]

False Negatives (FN): 14,943 = 7.3% of all positives. These are real transits the model missed. At 7.3%, this is an acceptable operating point for a first-pass screening tool, provided downstream BLS/radial velocity follow-up catches some of these. The asymmetry of cost (missed transit = lost science; false positive = caught by RV follow-up) justifies this operating point.

False Positives (FP): 126,784 = 13.3% of all negatives. Approximately 1 in 7.5 negative windows is incorrectly flagged as a transit. This is the dominant source of noise in the pipeline output. It is a direct consequence of setting the threshold at 0.69 to achieve 92.7% recall. The precision-recall tradeoff here is explicit and deliberate.

True Negatives (TN): 827,306. The model correctly rejects 86.7% of negative windows.

The row-normalised version of this matrix (which the notebook generates) would show: 92.7% recall on positives, 86.7% specificity on negatives.

Figure 3: Precision–Recall Curve (Val Set)

What you see

A PR curve plotted from threshold 0.01 to 0.95, with the operating point at $t=0.69$ marked. The curve starts at high recall/low precision (bottom-right) and sweeps to high precision/low recall (top-left). The operating point at $t=0.69$ sits at approximately (Recall=0.9275, Precision=0.6011).^[1]

Interpretation

Shape of the curve: The PR curve shows a characteristic pattern for an imbalanced problem with a strong classifier — a broad, high plateau of recall that stays above 0.95 for thresholds from 0.01 to 0.72, then a sharp dropoff above $t=0.72$. This means the model has learned very confident representations for most transits, with a long tail of borderline cases.

The precision ceiling at ~0.60–0.62. Even at the best achievable F1, precision does not exceed ~0.72 (at high threshold, low recall). This is the critical limitation of this run. There are three structural reasons for this:

1. **Label noise:** The KOI-based labeling uses a $0.5\times$ duration buffer, so windows near the transit edge that are labeled negative might contain partial transit signals, creating systematically ambiguous examples near the decision boundary.
2. **Secondary eclipse confusion:** The secondary eclipse channel, by design, produces a dip at phase 0.5. However, the model is now classifying some windows where the secondary channel shows a dip (i.e., negative windows where the light curve coincidentally dips around phase 0.5) as positive. This is a hypothesis worth testing.
3. **~1:4 class imbalance in the evaluated sets:** Even a 5% base-rate confusion probability on negatives generates many FPs.

Discontinuity at $t=0.54$. The sweep data shows an abrupt precision jump from 0.5421 at $t=0.535$ to 0.5787 at $t=0.540$ — a gain of 0.037 in precision with only a 0.014 drop in recall. This is not noise. It indicates a **tight cluster of borderline examples** with sigmoid outputs concentrated just below 0.54. These are likely the most ambiguous windows — low-signal transits or secondary-eclipse-dip windows that the model is almost classifying correctly, all concentrated at a particular probability value.^[1]

The manually selected operating point at $t=0.69$: The automatic sweep selected $t=0.74$ ($P=0.621$, $R=0.859$, $F1=0.721$). The threshold was manually overridden to $t=0.69$ ($P=0.601$, $R=0.9275$, $F1=0.7295$). This is a deliberate science decision: trading 3.5 points of precision for 6.8 points of recall. For a blind survey where you cannot afford to miss transits, this is the correct direction.^[1]

Figure 4: ROC Curve (Val Set)

What you see

The ROC curve shows True Positive Rate vs. False Positive Rate across all thresholds. The operating point at $t=0.69$ is marked. $AUC = 0.9429$.^[1]

Interpretation

$AUC = 0.9429$. This is a strong result. To contextualise: the original Astronet paper (Shallue & Vanderburg 2018) reported $AUC \sim 0.988$, but on a cleaner dataset with longer windows and global view + local view dual-input structure. For a first dual-channel InceptionResNet1D run on the more challenging windowed sliding-window framing, 0.9429 is competitive.

Operating point location. At $t=0.69$, the operating point sits at approximately $FPR=0.133$, $TPR=0.9275$ on the ROC curve. The curve's steepness before this point (low FPR, high TPR) is what you want — it means the model is highly confident about most true positives and only starts accepting false positives as you push toward higher recall.

Comparison to the PR curve. The ROC curve looks more favourable than the PR curve ($AUC=0.943$ vs $AP \approx 0.65\text{--}0.70$ estimated). This discrepancy is expected under class imbalance — ROC is insensitive to base rate, while PR curves expose the difficulty of maintaining precision when positives are a minority. The PR curve is the more honest diagnostic for this use case.

Final Numbers: Val vs. Test

Metric	Validation (t=0.69)	Test (t=0.69)
Precision	0.6011	0.6095
Recall	0.9275	0.9324
F1	0.7295	0.7371
AUC-ROC	0.9429	0.9415
Accuracy	0.8778	0.8791
TP	191,072	196,785
FP	126,784	126,093
FN	14,943	14,265
TN	827,306	824,163
Missed transits	7.3%	6.8%
False alarms	13.3%	13.3%

What this means

The val-test consistency is excellent. Test recall is **slightly higher** than val (93.2% vs 92.8%), test precision is **slightly higher** (60.9% vs 60.1%), and test AUC is **marginally lower** (0.9415 vs 0.9429). All differences are within 0.7 percentage points. There is no evidence of val-set overfitting through threshold selection. The model generalises cleanly. The FP rate is identical at 13.3% on both splits, which is particularly reassuring — it means the false alarm rate is a structural property of the model, not a function of the specific val windows used for threshold selection.^[1]

What Should Happen Next

There are five prioritised diagnostic and improvement directions, in order of urgency:

- 1. Investigate the precision ceiling.** Precision capping at ~0.60 is the dominant limitation. The first step is to audit the FP windows — load the 126K val false positives, plot channel 0 and channel 1 for a random sample, and look for patterns. If most FPs show near-flat channel 0 but a noisy channel 1 dip, secondary eclipse confusion is the culprit and the architecture needs a channel-attention gate. If FPs look like genuine flux variations (stellar spots, binary EBs), the label noise hypothesis is dominant and the KOI labeling strategy needs refinement.
- 2. Investigate the epoch 76 spike.** Load `epoch076.pt` and compare its weights to `epoch065.pt`. If the weight delta is large in specific layers (e.g., the SE attention weights or the first Inception block), you've found which part of the network is unstable. A gradient clipping tightening from 1.0 to 0.5, or a reduction in `pos_weight` from 4.2 to 3.5, would reduce this instability in the next run.

3. Single-channel ablation. You do not yet have empirical evidence that the secondary eclipse channel is helping. The correct comparison is to run the same training with `in_channels=1` on the original windows/ data at otherwise identical settings. If AUC drops below 0.93, the secondary channel is contributing; if it does not, the channel is not yet providing useful signal and the preprocessing for channel 1 needs revisiting.

4. Adjust posweight. The first two epochs triggered all-positive collapse with `pos_weight=4.2`, and epoch 76 triggered it again. Consider reducing to 3.5 (still 11% above natural ratio 3.15), which would reduce the collapse risk while maintaining the recall bias.

5. Threshold at 0.69 vs. 0.74. The manual choice of $t=0.69$ over the algorithmic $t=0.74$ is scientifically justified but should be documented with an explicit science case. If this model feeds into a BLS period search pipeline, the false positive load at $t=0.69$ (13.3%) vs. $t=0.74$ ($\approx 11\%$) matters for downstream compute cost. Quantify the gain in recovered planets (the $\Delta FN = 14,943$ at $t=0.69$ vs. $\sim 17,500$ estimated at $t=0.74$) against the downstream processing load before committing to the operating threshold.

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1. 2026-04-11-Dual-Channel-InceptionResNet-Model-Building.ipynb